

National Aeronautics and
Space Administration



Synthetic transcriptomics for mouse spaceflight

Can generated RNA-seq improve FLT vs GC
analysis within each tissue?

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QUESTION

Small studies and study effects complicate tissue comparisons

Can a generator help rank spaceflight signal without pretending it creates new animals?

NASA OSDR

1,610

biological profiles

75

accessions

835 FLT

775 GC

ARCHS4 mouse

997,515

profiles audited

17,244

selected

20 tissues

- Mission, strain, material and assay differences can resemble a flight effect.
- The generator must preserve tissue and FLT/GC structure without memorizing samples.
- Statistical evidence must still come from observed OSDR profiles.

We built a configurable bulk RNA-seq generation pipeline

Each stage exposes alternatives; a strong outline marks the configuration used downstream.

1

Data scope

Sources

• OSDR API

• ARCHS4

Studies

Single

• Multiple

Tissues

Per tissue

• All tissues

2

Transform

Expression

Raw

CPM

• TPM

Scaling

None

Z-score

• MaxAbs

Features

All

HVG

• L1000

3

Harmonization

Global or study correction

• None

Within-study z-score

ComBat / MBatch

MOBER

4

Model training

Generator

WGAN-GP

• DDIM

Training source

OSDR only

ARCHS4 only

• Pretrain + adapt

5

Conditioning

Model inputs

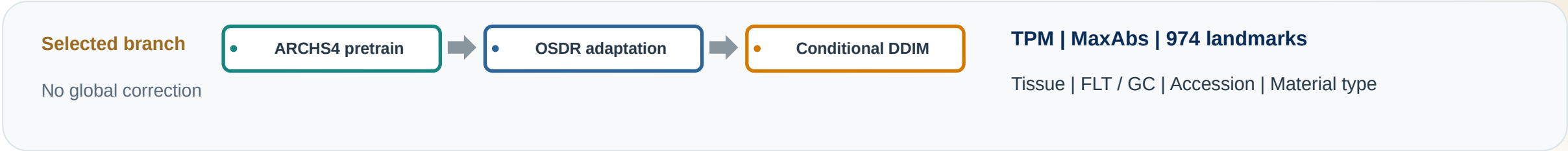
• Tissue

• FLT / GC

• Accession

• Material type

Sex / age



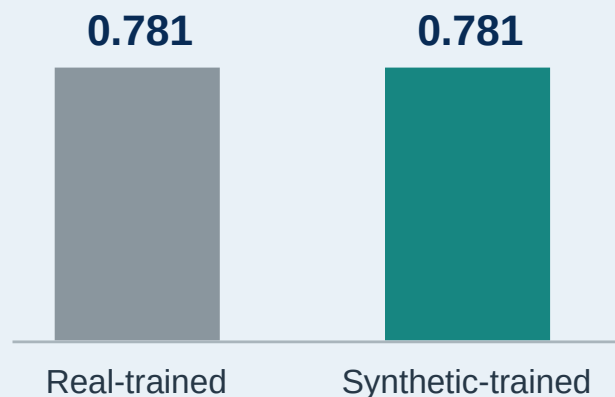
VALIDATION

DDIM matched expression and reduced separability

AA closer to 0.5 means a classifier has less success separating real and generated profiles; lower FD is better.

ARCHS4 tissue probe

Balanced accuracy on held-out real profiles



Metric	WGAN-GP	DDIM	Reading
Correlation	0.976	0.974	similar
F1	0.985	0.997	higher
Adversarial accuracy	0.636	0.475	closer to 0.5
FD / real P95	0.144	0.074	lower

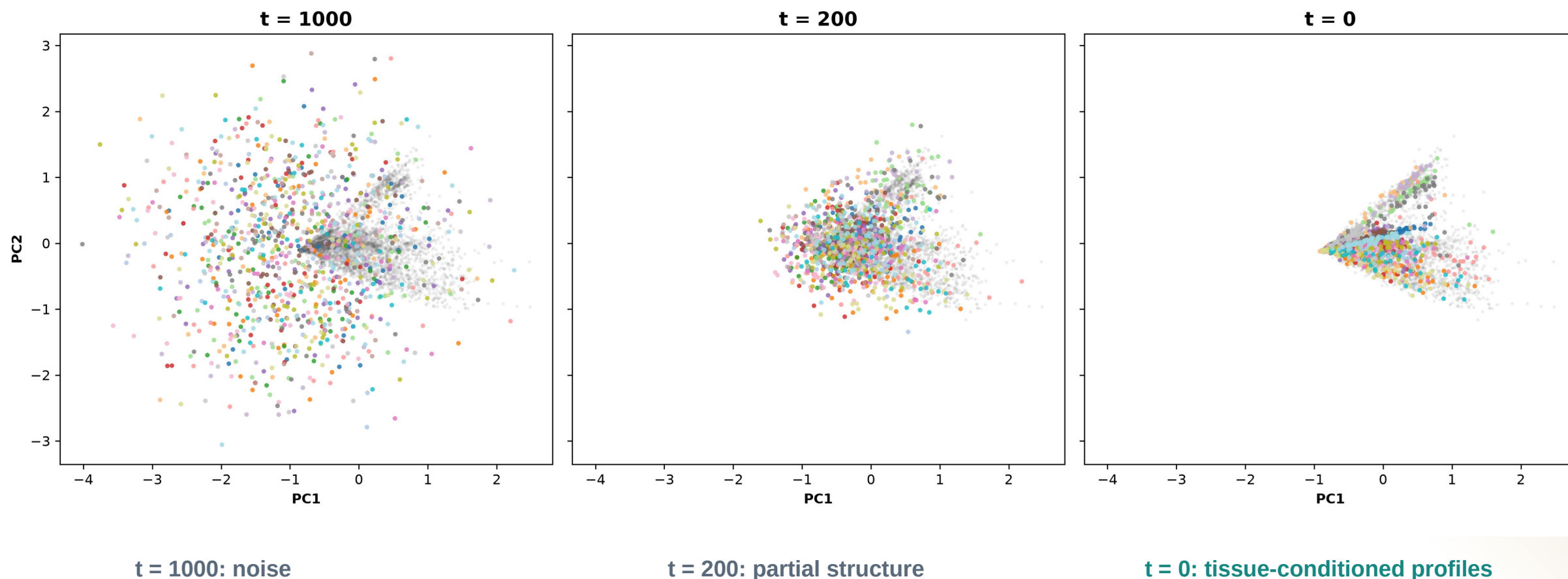
Use DDIM downstream

Lower separability and distributional distance, with comparable correlation and higher F1.

DIFFUSION

Diffusion learns tissue structure from noise

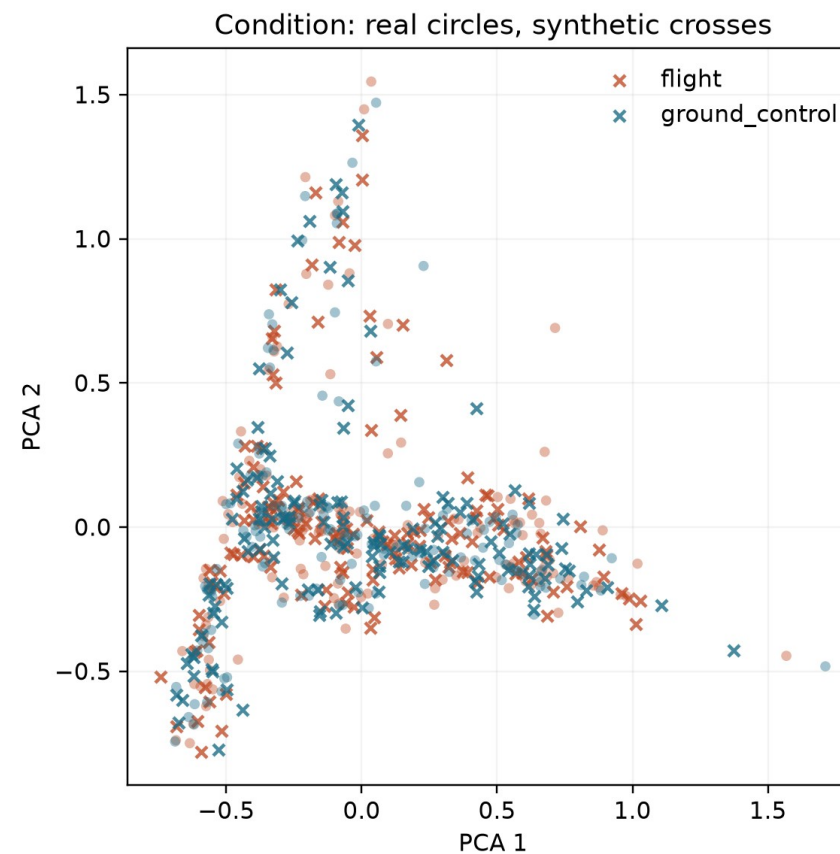
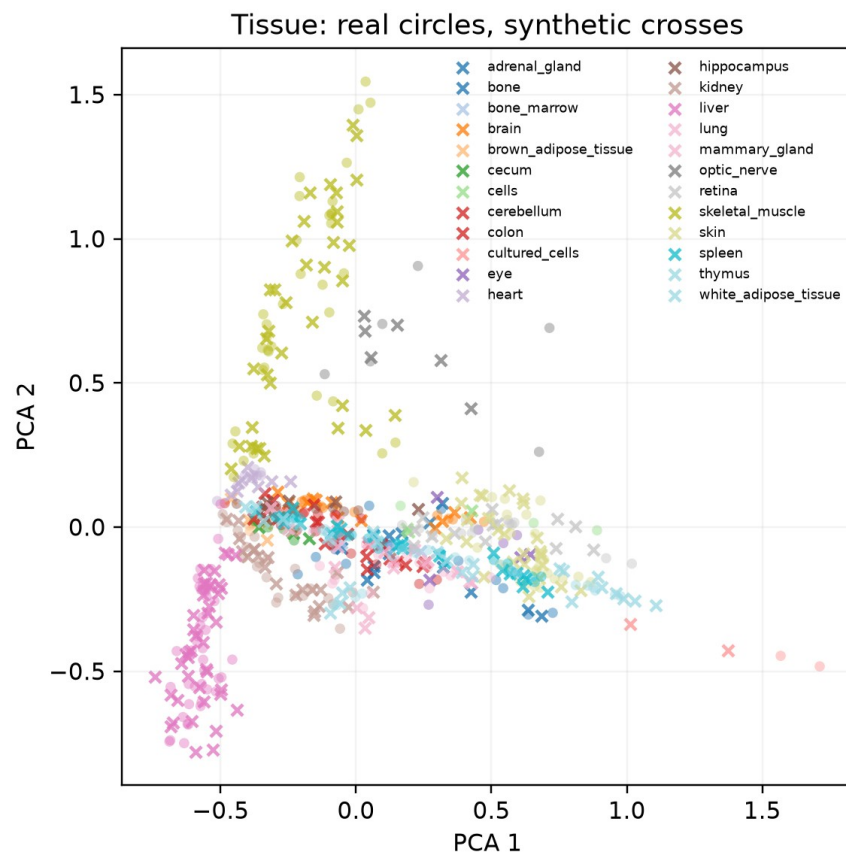
The same PCA axes follow 1,024 generated profiles through reverse diffusion.



DIFFUSION OUTPUT

Generated profiles track the real OSDR PCA manifold

Circles are locked real OSDR profiles; crosses are matched DDIM profiles from generation seed 5020.

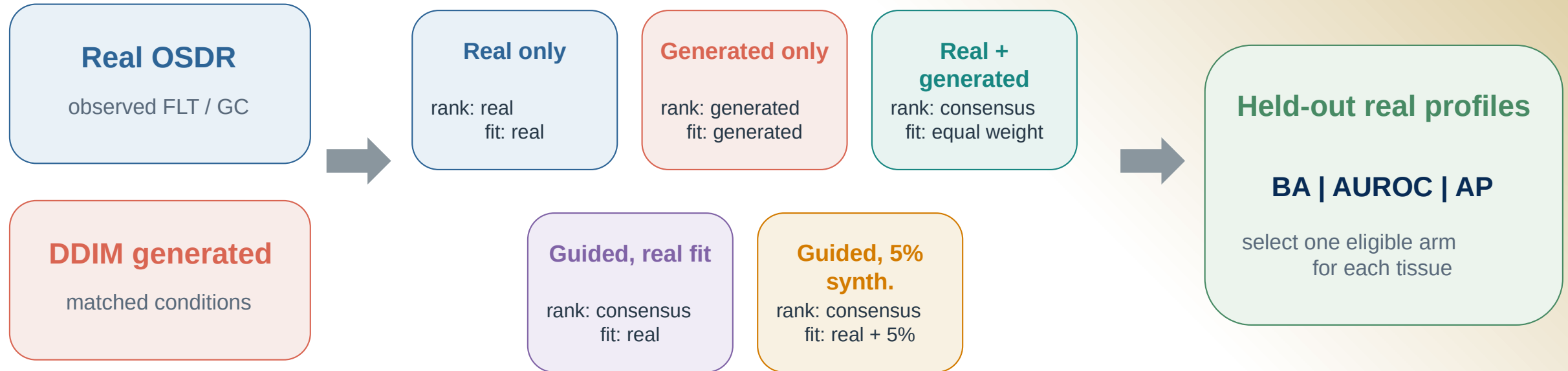


Interpretation: tissue structure dominates; FLT and GC are subtler. PCA is descriptive, so quantitative metrics determine validation.

ANALYSIS

Synthetic profiles entered the analysis in five different ways

Every arm was judged on held-out real profiles before synthetic attribution was allowed.



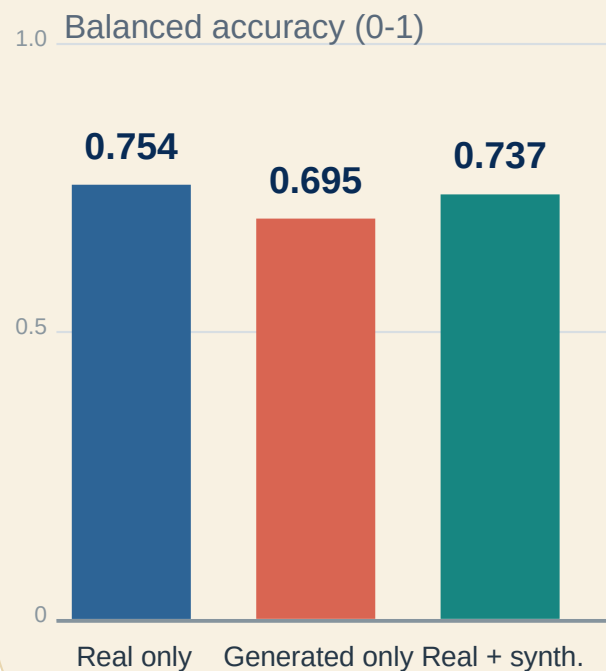
Biology check FLT vs GC effects, random-effects meta-analysis and BH FDR use real OSD R profiles only.

Synthetic profiles never increase animal n.

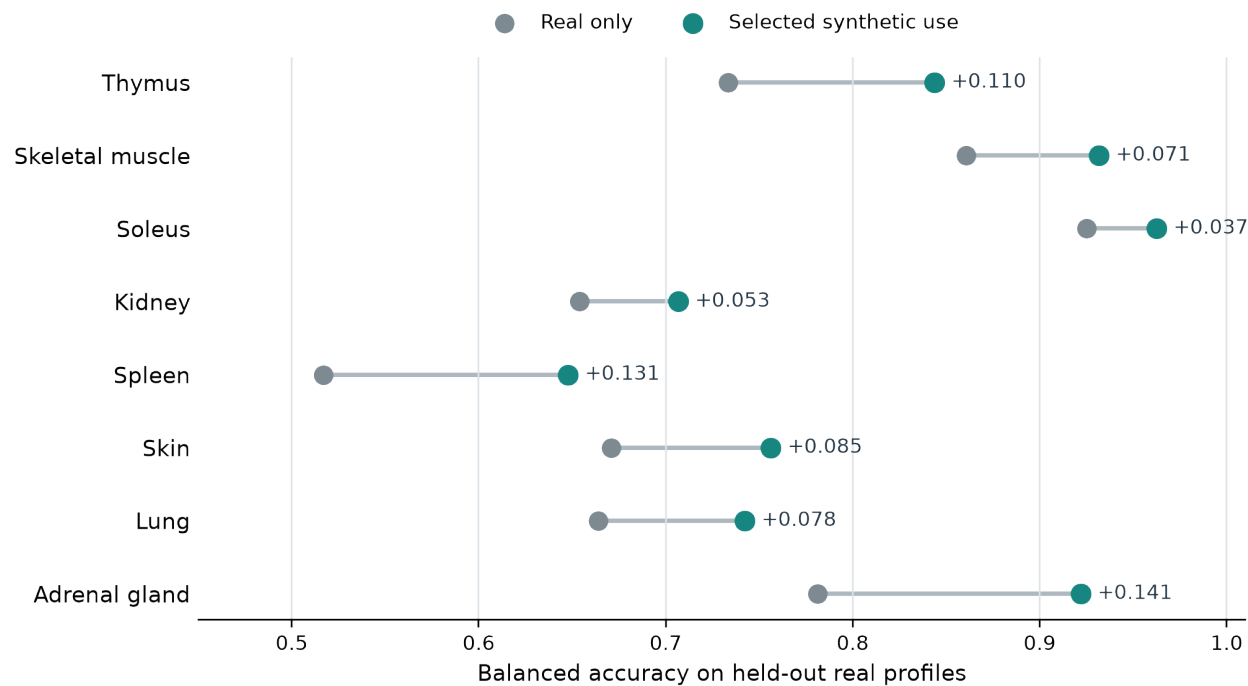
Pooling tissues hid useful signal

A single FLT/GC classifier did not improve, but tissue-specific synthetic use often did.

Pooled across tissues



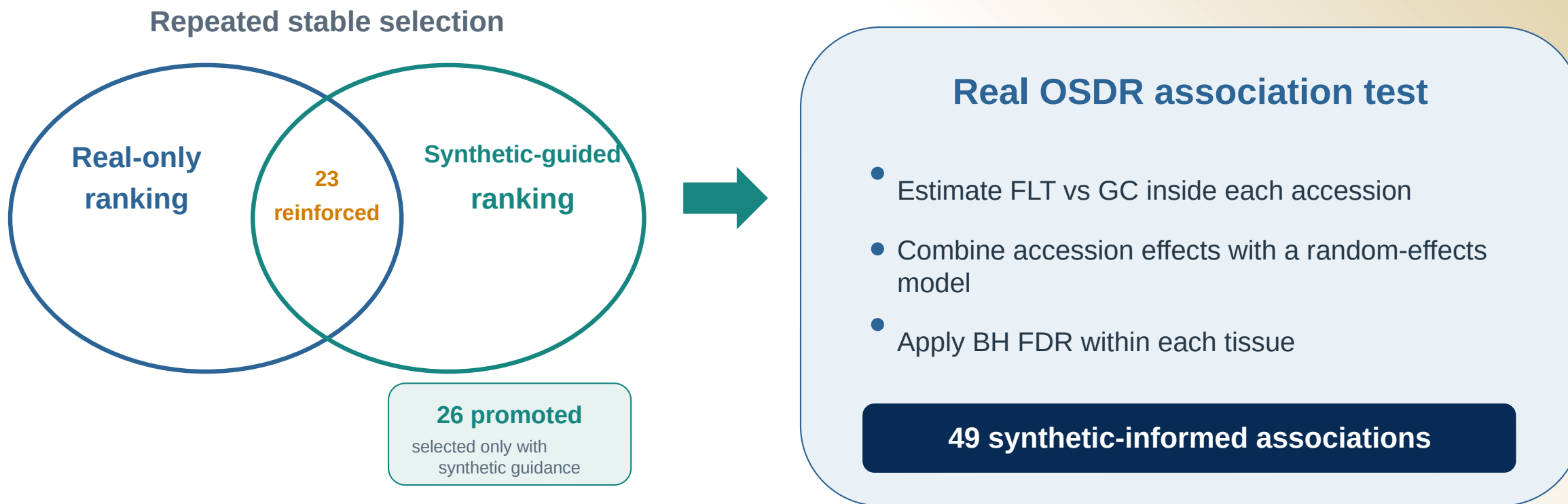
Selected use within each tissue



The useful arm differed by tissue.

Synthetic guidance changed ranking, not statistical evidence

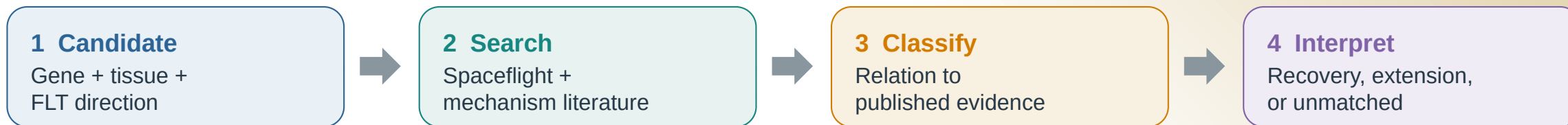
Promoted and reinforced describe repeated feature selection; both require a supporting effect in real OSDR data.



These are 49 tissue-gene associations among 459 BH-FDR rows. A gene can appear in more than one tissue.

Selection and literature are separate dimensions

All 49 synthetic-informed associations received both a selection status and a literature interpretation.



Selection status

26

Promoted

Stable only with synthetic guidance

23

Reinforced

Stable with and without guidance

Literature interpretation

22

Aligning

Direct or same-tissue process agreement

19

Complementary

Related process or mechanism

4

Ambiguous

Mixed or context-dependent evidence

4

Unmatched

No sufficiently specific match found

A reinforced gene can be complementary, ambiguous, or unmatched; an aligning gene can be promoted.

ANALYSIS COVERAGE

All 27 completed tissue analyses were retained

The biological narrative narrows only after reporting synthetic-informed, real-only, and null outcomes.

10

Synthetic-informed

Promoted or reinforced BH-FDR gene

Adrenal Gland
Eye
Kidney
Skeletal muscle (pooled)
Skin
Spleen
Thymus
Gastrocnemius
Soleus
Tibialis Anterior

5

Real BH FDR only

Real association, no synthetic-informed selection

Heart
Liver
Retina
EDL
Quadriceps

12

No panel BH-FDR gene

No BH-FDR gene in the 974-gene panel

Bone
Bone Marrow
Brain
Brown Adipose Tissue
Cecum
Cerebellum
Colon
Hippocampus
Lung
Mammary Gland
Optic Nerve
White Adipose Tissue

GENE INVENTORY

Synthetic-informed genes spanned 10 tissue analyses

All 49 associations passed BH FDR in real OSD data; synthetic profiles affected prioritization, not the statistical test.

[P] promoted [R] reinforced aligning complementary ambiguous unmatched

Tag = selection | Gene color = literature

Thymus 16 genes	FLT higher	[P]Hsd17b11, [P]Etv1, [R]Snx7
	FLT lower	[P]Nusap1, [P]Stmn1, [P]Birc5, [P]Cdk1, [P]Top2a, [P]Ccnb2, [P]Aurka, [P]Ccne2, [P]Kif20a, [P]Pcna, [P]Ccnf, [R]Ube2c, [R]Gmnn

Spleen 4 genes	FLT higher	[P]Rai14, [P]Myl9, [P]Ptprk, [R]Loxl1
	FLT lower	none

Kidney 2 genes	FLT higher	[P]Inpp4b, [R]Slc37a4
	FLT lower	none

Gastrocnemius 2 genes	FLT higher	[P]Nfkb1a
	FLT lower	[P]Fhl2

Eye 1 gene	FLT higher	none
	FLT lower	[R]Klhl21

Skeletal muscle (pooled) 12 genes	FLT higher	[R]Sox4, [R]Cebpd, [R]Sh3bp5, [R]Prkcd, [R]Arid5b, [R]Sesn1, [R]Tle1
	FLT lower	[P]Klhl21, [P]Mapkapk5, [P]Reep5, [P]Itgb5, [R]Bphl

Soleus 5 genes	FLT higher	[R]Tpm1
	FLT lower	[R]Bdh1, [R]Ech1, [R]Bnip3, [R]Decr1

Tibialis anterior 4 genes	FLT higher	[P]Cebpd, [R]Cdkn1a, [R]St3gal5, [R]Bnip3
	FLT lower	none

Adrenal gland 2 genes	FLT higher	none
	FLT lower	[P]Psm8, [R]Tspan4

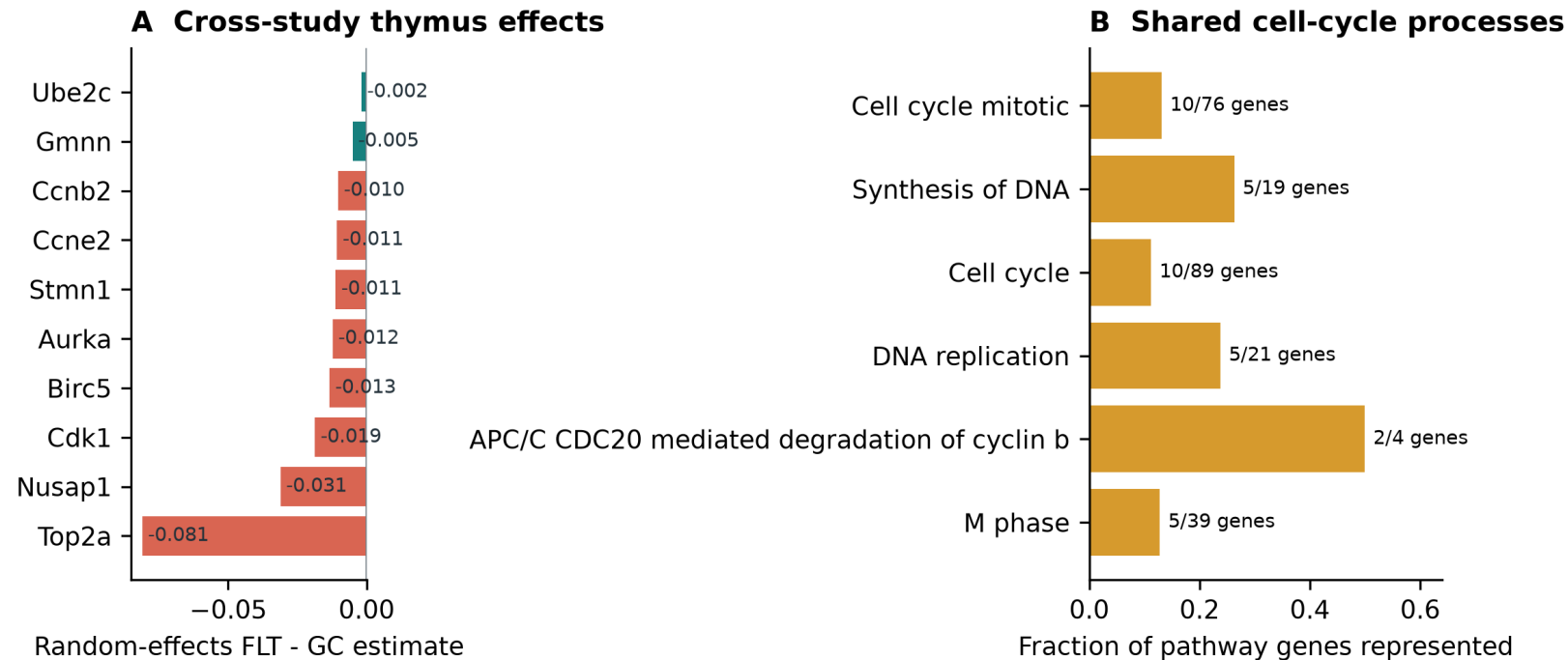
Skin 1 gene	FLT higher	[P]Plscr1
	FLT lower	none

THYMUS

Thymus points to lower proliferative renewal

The core panel recovers cell-cycle biology; two flight-higher genes extend the hypothesis.

Synthetic-informed thymus genes converge on a flight-lower mitotic program



16 synthetic-informed
BH-FDR associations

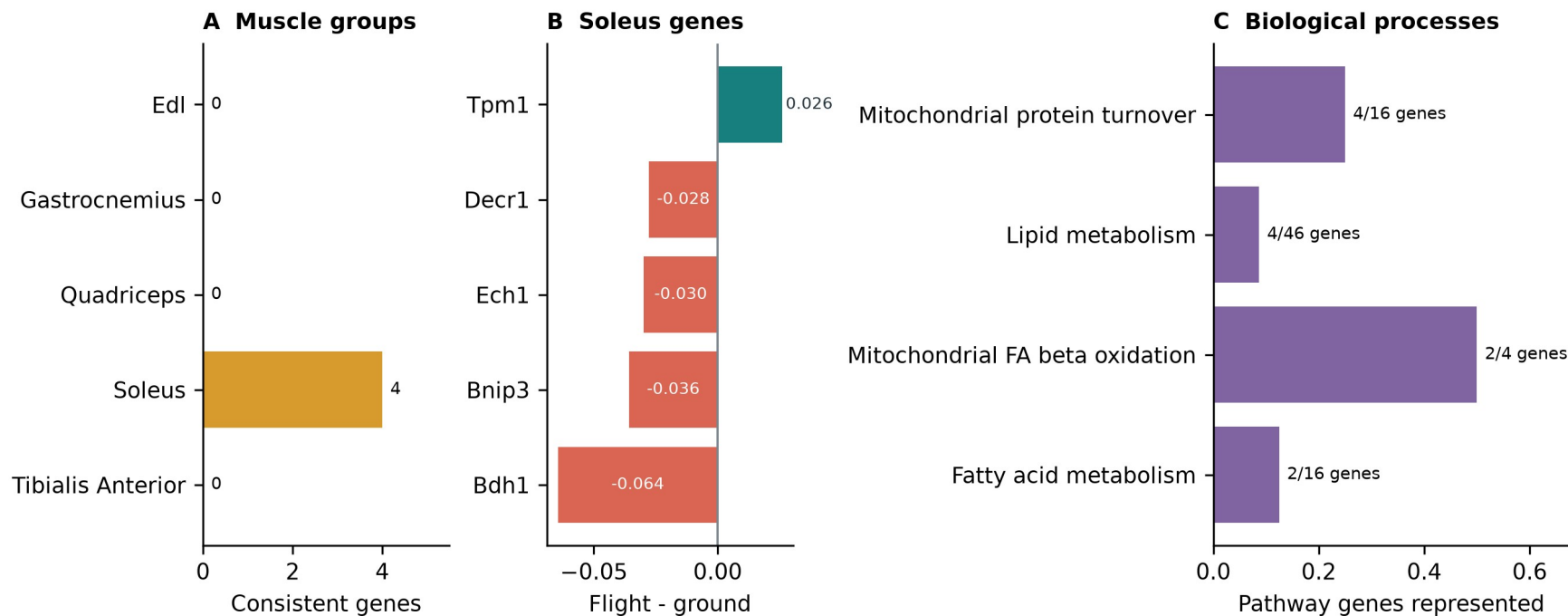
13 promoted | 3 reinforced

- Lower Nusap1, Stmn1, Birc5, Cdk1, Top2a and related genes
- Cell cycle, DNA replication, APC/C and G2/M processes
- Higher Hsd17b11 and Etv1 suggest lipid or T-cell-state shifts
- Bulk RNA-seq cannot separate cell loss from lower transcription

Soleus reinforces a mitochondrial and lipid program

This result strengthens an existing real-data pattern rather than promoting a new soleus gene.

Anatomical separation reveals a soleus-specific oxidative-metabolism hypothesis



0.925 -> 0.963

balanced accuracy

5 reinforced | 0 promoted

- Lower Bdh1, Ech1, Bnip3 and Decr1
- Higher Tpm1
- Mitochondrial turnover and fatty-acid metabolism
- Compatible with unloading and contractile remodeling

ADDITIONAL FINDINGS I

Each additional tissue defines a separate hypothesis

Promoted and reinforced selections are separated within every tissue.

Analysis unit	Selection	Genes and FLT direction	Interpretation
Pooled muscle	Promoted	Lower: Klhl21, Mapkapk5, Reep5, Itgb5	Heterogeneous stress, differentiation, interferon and sialic-acid response; interpret with the anatomical muscle groups.
	Reinforced	Higher: Sox4, Cebpd, Sh3bp5, Prkcd, Arid5b, Sesn1, Tle1; lower: Bphl	
Kidney	Promoted	Inpp4b higher	Renal phosphoinositide signaling and glucose-handling hypothesis; the pair had no shared Reactome enrichment.
	Reinforced	Slc37a4 higher	
Spleen	Promoted	Rai14, Myl9, Ptprk higher	Adhesion, actomyosin and extracellular-matrix or immune-organization hypothesis; no coherent pathway enrichment.
	Reinforced	Loxl1 higher	
Skin	Promoted	Plscr1 higher	Interferon-linked skin candidate within broader cell-cycle and DNA-repair responses; a single-gene result.
	Reinforced	None	

ADDITIONAL FINDINGS II

Eye, adrenal and muscle-group results remain tissue-specific

Promoted and reinforced selections are separated within every tissue.

Analysis unit	Selection	Genes and FLT direction	Interpretation
Eye	Promoted	None	Process-level alignment with lower proliferation and cytokinesis in flight eye tissue; not an exact prior gene replication.
	Reinforced	Klhl21 lower	
Adrenal gland	Promoted	Psmb8 lower	Literature-unmatched immune, proteostasis or tissue-composition candidates; no adrenal mechanism is established.
	Reinforced	Tspan4 lower	
Gastrocnemius	Promoted	Nfkb1a higher; Fhl2 lower	NF-kappaB stress alignment plus an autophagy or myogenesis candidate; the two genes do not form a coherent pathway.
	Reinforced	None	
Tibialis anterior	Promoted	Cebpd higher	Stress, cell-cycle, ganglioside-signaling and mitophagy candidates with mixed or complementary prior evidence.
	Reinforced	Cdkn1a, St3gal5, Bnip3 higher	

TAKEAWAYS

Synthetic data helped most as a tissue-specific prior

The generator created useful structure, not new biological replication.

1 Generate

Conditional DDIM produced high-fidelity profiles with near-chance real-versus-synthetic separation.

2 Use

Pooled augmentation failed. Tissue-specific ranking and low-weight training were more useful.

3 Interpret

Thymus and soleus were clearest. Literature review separated recovery from complementary hypotheses.

Next test

Use independent samples and cell-resolved assays to test thymus proliferation, soleus metabolism and prioritized candidates from additional tissues.

Generated profiles never enter the biological sample count or the BH-FDR test.

Thank you

Questions?

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Mentor

James Casaletto

Program

NASA Space Life Sciences Training Program

Data

NASA OSDR | ARCHS4 | Reactome

Compute

NASA Ames Research Center

Code and
manuscript

github.com/jasont314/nasa-mouse

All biological associations were tested in observed OSDR profiles.